Determinants of Household Participation in Securities Markets: An Empirical Analysis

Shivansh Verma
Dept of CS
Ashoka University
Sonipat, India
shivansh.verma_ug2023@ashoka.edu.in

Sashwat Dhanuka
Dept of CS
Ashoka University
Sonipat, India
sashwat.dhanuka_ug2023@ashoka.edu.in

***Abstract*—Household participation in capital markets is a crucial driver of financial inclusion and wealth creation, yet it remains disproportionately skewed across demographic lines. Using a large-scale micro-dataset from the 2025 SEBI Household Survey ($N = 109,430$), this paper empirically analyzes the determinants of securities market participation in India. We employ a weighted logistic regression framework with HC3 robust standard errors across twelve predictors spanning awareness, socioeconomic, demographic, and geographic dimensions, revealing that product awareness is the single strongest driver of participation (OR = 2.22, $p < 0.001$), exceeding income (OR = 1.60) and education (OR = 1.56), while two predictors (Gen X life stage and West zone) return null results, validating model specificity. The model achieves a McFadden pseudo-R^2 of 0.290, confirming that awareness is an independent, policy-actionable determinant beyond income and education alone. We further investigate structural barriers deterring non-investors, identifying loss aversion and information asymmetry as primary friction points. To deepen our behavioral insights, we expose a stark Dunning-Kruger effect among overconfident retail traders and measure the impact of "finfluencers" on high-risk asset adoption. Finally, we implement Histogram-based Gradient Boosting Classifiers to predict participation, asset choice, and investment horizons. Our predictive models achieve strong discriminatory power for baseline participation (AUC = 0.814) and asset allocation (AUC = 0.820), validating that while market entry is structurally gated, post-entry behavior is heavily dictated by psychographic traits and financial literacy.**

I. Introduction

The democratization of capital markets is essential for broad-based economic growth and household wealth accumulation. In emerging economies like India, retail participation in securities has witnessed a significant surge, catalyzed by digitization and the proliferation of zero-commission trading platforms. Despite this growth, participation remains highly heterogeneous.

Understanding the determinants of household participation is a central problem in financial economics. Why do certain households allocate capital to mutual funds and equities, while others restrict their savings to traditional, low-yield instruments like fixed deposits? The literature points to a combination of background risk, financial literacy, and trust in institutions. However, most existing studies rely on older data that predate the post-pandemic retail boom, the widespread adoption of digital KYC, and the systemic rise of social media financial influencers.

This paper addresses this gap by utilizing the comprehensive 2025 SEBI Household Survey. We pose four core research questions: (1) Who is likely to participate in the securities market? (2) What structural and behavioral barriers keep the majority of households out? (3) Which securities do participating households prefer, and for how long? and (4) Can we predict these outcomes given basic demographic and behavioral metrics?

The remainder of the paper is organized as follows: Section II details the dataset and methodology. Section III presents the empirical findings regarding determinants, behavioral anomalies, and securities preferences. Section IV details our predictive modeling framework. Finally, Section V concludes with highly condensed policy implications.

II. Data and Methodology

A. Dataset

We utilize micro-data from the 2025 SEBI Household Survey on Securities Market Participation, a nationally representative cross-sectional survey of $N = 109{,}430$ households administered across four geographic zones: North, South, East, and West India. Each observation corresponds to a unique household respondent, weighted for population representativeness via a pre-computed survey weight variable.

B. Data Normalization and Schema Architecture

To facilitate robust multivariate analysis, the raw survey microdata—originally comprising a single flat file with 449 columns—underwent a rigorous Extraction, Transformation, and Loading (ETL) process. Categorical survey responses and qualitative inputs were systematically transformed into strictly analytical data types (booleans, integers, and ordinal scales), eliminating unstructured free-text to ensure computational efficiency and statistical reliability.

As illustrated in Fig. 1, the transformed data was subsequently loaded into a fully normalized PostgreSQL relational database. The schema architecture is anchored by a central `respondents` table, which establishes a unique primary

Based on the 2025 SEBI Household Survey on Securities Market Participation.

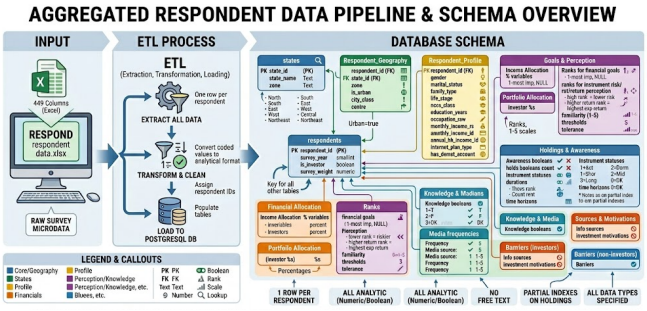

Fig. 1. Architecture of the data layer showing the aggregation and normalization process of the respondent microdata.

key (`respondent_id`) for each household and stores the population-representative survey weights.

The remaining data is segmented across 16 interconnected, domain-specific tables encompassing geographic distribution, socioeconomic profiles, financial allocations, product awareness, and behavioral perceptions. Maintaining a strict one-to-one relationship with the core identity table, this modular and highly indexed design allows for rapid, optimized querying of specific behavioral intersections—such as isolating active equity investors to correlate objective financial literacy with risk tolerance—thereby forming the computational backbone for both our logistic regression and gradient boosting frameworks.

The sample is bifurcated into *active investors* (those holding at least one financial security) ($n = 27{,}882$; 25.5%) and *non-investors* ($n = 81{,}548$; 74.5%). Table I reports key demographic characteristics. The sample skews male (64.5%) and is concentrated in younger cohorts: Gen Z (18–27 years) and Millennials (28–43 years) together comprise 86.1% of respondents. Nuclear family structures predominate (84.2% of households).

TABLE I
SAMPLE COMPOSITION AND PARTICIPATION RATES

	N	% of Sample	Participation Rate
Gender			
Male	70,616	64.5	30.8%
Female	38,814	35.5	15.8%
Life Stage			
Gen Z (18–27)	49,035	44.8	24.3%
Millennial (28–43)	45,235	41.3	28.1%
Gen X (44–59)	12,806	11.7	22.9%
Baby Boomer (60+)	2,354	2.2	13.3%
Residence			
Urban	76,931	70.3	28.8%
Rural	32,499	29.7	17.6%
Zone			
West	34,153	31.2	27.1%
South	27,885	25.5	26.8%
North	26,439	24.2	24.1%
East	20,953	19.1	22.8%

The survey employs a conditional routing design. Investors were directed to dedicated sub-modules (SS_B3, SS_B10) covering information sources and investment motivations, yielding $n = 9{,}797$ complete investor sub-responses. Non-investors were routed to a separate module (SS_BB2) eliciting up to three self-reported barriers to participation, yielding $n = 14{,}794$ barrier responses. All group-level percentages for these modules are computed against the respective routing-eligible population, not the full sample.

C. Exploratory Analysis

The raw participation patterns in Table I reveal a market sharply stratified by socioeconomic position. The gender gap is the most visually immediate: male respondents participate at 30.8% against 15.8% for female respondents. The income gradient, however, is the steepest univariate signal: participation rises from 10.7% in the lowest quintile (below INR 12,500/month) to 47.6% in the highest (above INR 35,000/month), a more than fourfold difference. Urban households (28.8%) participate at nearly double the rate of rural ones (17.6%). Notably, Millennials (28–43 years) record the *highest* participation rate of any cohort (28.1%), exceeding Gen X (22.9%) and Gen Z (24.3%), suggesting the post-2020 retail boom was disproportionately captured by this age group rather than younger entrants.

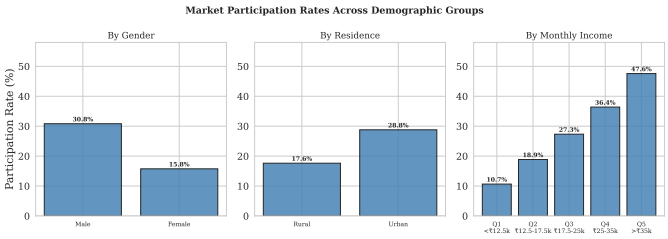

Fig. 2. Market participation rates by gender, residence, and monthly income quintile. The income gradient (10.7% to 47.6%) is the steepest univariate signal, while the gender gap (30.8% vs 15.8%) persists across all income levels.

A critical precondition for participation is product awareness. The survey's awareness module reveals a steep gradient: fixed deposits are known to 99.1% of respondents, while mutual funds and ETFs reach 67.0%, direct equity 62.0%, and derivatives just 17.9%. This awareness gap, between near-universal familiarity with traditional instruments and niche awareness of market-linked ones, maps closely onto the observed holdings distribution. Among the 27,882 active investors, MF/ETFs lead penetration (66.5%), followed by FD/RD (57.2%) and direct equity (49.3%). The co-existence of MFs and FDs as the two most-held instruments indicates that Indian retail investors are not homogenously risk-seeking; the portfolio evidence points instead to a layered allocation model where market-linked products supplement rather than replace traditional savings.

The financial literacy data deepens this picture. When presented with nine objective financial knowledge questions, investors perform unevenly: while 80%+ correctly identify that KYC can be completed online and that higher returns entail higher risk, a majority respond incorrectly or "Don't Know" to

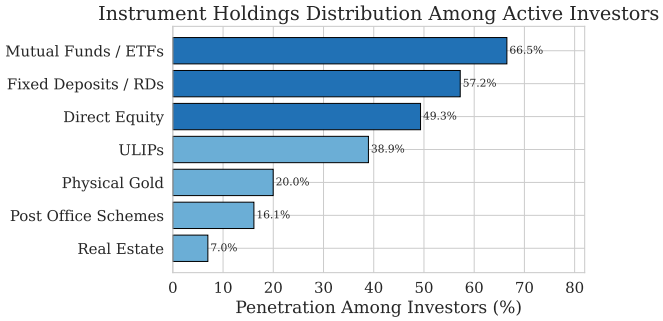

Fig. 3. Instrument holdings penetration among active investors ($n = 27{,}882$). MF/ETFs and Fixed Deposits co-dominate, indicating a layered allocation model rather than a shift away from traditional savings.

questions about compounding, diversification, and even basic SEBI tools (CAS, BSDA). This asymmetry (high procedural literacy, weak conceptual literacy) is directly relevant to the Dunning-Kruger analysis in Section III-A and to the barriers analysis, where "Lack of Knowledge" ranks as the second most-cited deterrent among non-investors.

D. Variable Operationalization

The dependent variable $Y_i \in \{0, 1\}$ indicates whether household i holds any financial security. We include twelve regressors across four groups. The *awareness* block contains *Product Awareness* ($n_products_aware$; count of financial instruments the respondent has heard of, 0–15, drawn from the Q21A multi-select module administered to all 109,430 respondents). The *socioeconomic* block contains *Education* (continuous years of formal schooling) and *Income* (natural log of self-reported monthly household income in INR). The *demographic* block contains *Urban* (binary; urban vs. rural residence), *Gender* (binary; male = 1), and *Risk Tolerance* (ordinal; 1 = low, 2 = medium, 3 = high). The *life stage* block uses Gen Z as the reference category, with binary indicators for Millennial (28–43 years), Gen X (44–59 years), and Baby Boomer (60+ years). The *geographic* block uses the East zone as reference, with binary indicators for North, South, and West. All continuous regressors are standardized ($\mu = 0$, $\sigma = 1$) prior to estimation so that OR magnitudes are directly comparable across predictors.

E. Econometric Specification

We estimate a binary logistic regression via maximum likelihood:

$$\Pr(Y_i = 1) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i)}} \tag{1}$$

where $\mathbf{x}_i$ is the vector of standardized regressors. Coefficients are exponentiated to yield odds ratios (ORs), interpreted as the multiplicative change in participation odds per one-standard-deviation increase in each predictor. Statistical significance is evaluated via Wald z-tests with heteroskedasticity-consistent (HC3) standard errors; survey weights are incorporated during estimation. We note that with $N \approx 100{,}000$, statistical significance is trivially achieved for any real effect—p-values are reported for completeness but the economically meaningful metrics are OR magnitude and confidence interval width. We include 12 regressors spanning awareness, socioeconomic, demographic, and geographic dimensions; two (Gen X life stage and West zone) return ORs statistically indistinguishable from 1.0, providing a falsifiability check on the model.

F. Predictive Framework

To operationalize the regression findings, we implement Histogram-based Gradient Boosting Classifiers for three distinct prediction tasks: (Model 1) the binary likelihood of market participation; (Model 2) whether an investor holds market-linked instruments (equity or mutual funds) vs. traditional instruments only; and (Model 3) the propensity to hold equity over a long-term horizon. All models are trained on an 80/20 stratified split and evaluated via the area under the ROC curve (AUC).

III. Empirical Results

A. Socioeconomic Determinants of Participation

Table II presents the full logistic regression results ($N = 97{,}626$; McFadden pseudo-$R^2 = 0.290$, excellent fit range 0.20–0.40). Ten of twelve predictors are statistically significant; Gen X (vs. Gen Z) and West zone (vs. East) are not, with ORs of 0.993 and 1.021 respectively, confirming the model does not return spurious significance at this sample size.

TABLE II
Logistic Regression: Determinants of Market Participation

Predictor	OR	95% CI	z	Sig.
Awareness				
Product Awareness	2.218	[2.166, 2.272]	65.2	***
Socioeconomic				
Log(Income)	1.603	[1.564, 1.643]	37.4	***
Education (yrs)	1.556	[1.518, 1.594]	35.6	***
Demographic				
Male	1.526	[1.490, 1.564]	33.9	***
Urban	1.455	[1.422, 1.489]	32.0	***
Risk Tolerance	1.170	[1.145, 1.196]	14.0	***
Life Stage (ref: Gen Z)				
Millennial	1.124	[1.098, 1.151]	9.8	***
Gen X	0.993	[0.969, 1.017]	−0.6	
Baby Boomer	0.907	[0.881, 0.934]	−6.6	***
Zone (ref: East)				
North	0.850	[0.825, 0.876]	−10.7	***
South	0.842	[0.816, 0.869]	−10.7	***
West	1.021	[0.989, 1.054]	1.3	

N = 97,626 McFadden pseudo-R^2 = 0.290 (excellent fit: 0.20–0.40)
*** $p < 0.001$. HC3 robust SEs. Survey-weighted GLM (Binomial).

As depicted in Fig. 4, product awareness (OR = 2.22) is the dominant driver of participation, substantially exceeding income (1.60), education (1.56), male gender (1.53), and urban residency (1.45). The magnitude gap is meaningful: moving one standard deviation up in awareness predicts a

larger change in participation odds than any structural or demographic variable in the model.

The partial mediation structure is evident in the OR magnitudes: income (1.60) and education (1.56) retain large independent effects even after awareness is controlled, yet the relatively modest gap between them and awareness (2.22) indicates that part of the socioeconomic advantage operates through knowing more products. Awareness is therefore not merely a socioeconomic proxy; it is a distinct, policy-actionable lever that educated, higher-income households happen to possess more of.

The life stage results are instructive: Millennials participate *more* than Gen Z (OR = 1.12), plausibly reflecting the post-2020 retail boom; Gen X is statistically indistinguishable from Gen Z; and Baby Boomers participate significantly less (OR = 0.91). On geography, both South (OR = 0.84) and North (OR = 0.85) zones underperform East after controls, despite South's higher raw participation rates, indicating that South's raw advantage is fully explained by its demographic composition. West is no different from East after controls.

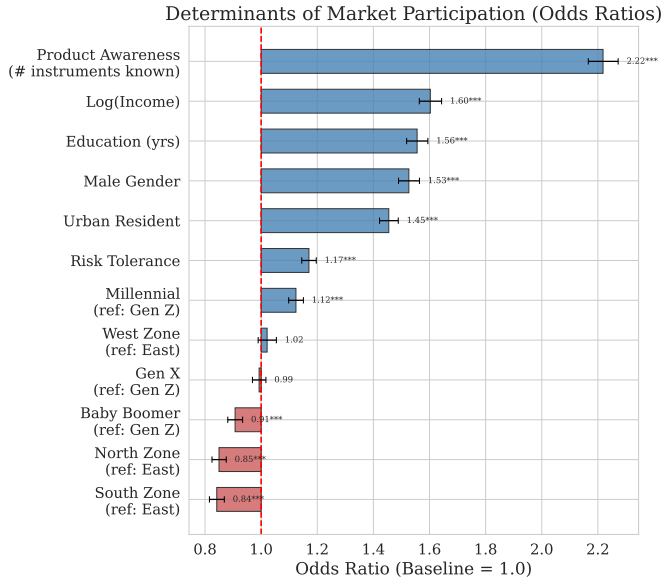

Fig. 4. Forest plot of odds ratios for determinants of market participation. Baseline is 1.0. Product awareness ($n_products_aware$, OR = 2.22) is the single strongest predictor, followed by log-income (1.60) and education (1.56). All predictors standardised to unit-SD units.

B. Barriers to Entry: Latent Archetype Analysis

To better understand the 74.5% of households that remain completely outside the securities market, we move beyond simple frequency rankings and cluster the 18 self-reported barriers into latent archetypes. Using K-means clustering on the subset of non-investors who identified specific deterrents ($n = 14,705$), we identify three distinct profiles: *Trust-Deficient* (46.3%), *Knowledge-Gated* (29.0%), and *Fear-Driven* (24.8%).

The demographic cross-tabulation (Fig. **??**) reveals that the *Knowledge-Gated* profile is disproportionately concentrated in rural areas and among the lowest income quartile (Q1). In contrast, *Trust-Deficiency* remains a persistent barrier that scales with wealth, remaining high even in urban higher-income cohorts. This segmentation implies that financial inclusion cannot be solved with a one-size-fits-all awareness campaign; it requires targeted interventions—educational for the rural knowledge-gated, and institutional/regulatory for the urban trust-deficient.

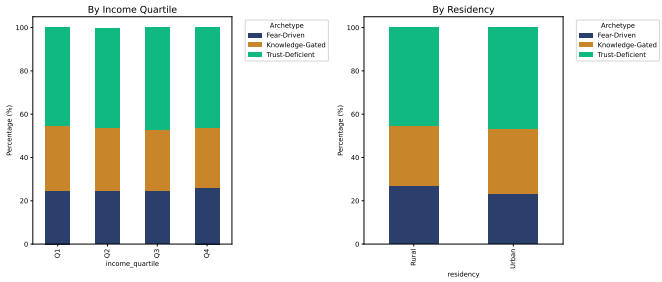

Fig. 5. Demographic distribution of barrier archetypes.

C. Behavioral Anomalies: Overconfidence & Dunning-Kruger

Diving deeper into the "Knowledge" barrier, we analyzed the divergence between self-reported market familiarity and actual scores on an objective financial literacy quiz. We discovered a profound Dunning-Kruger effect: investors who are "overconfident" (high perceived familiarity, low actual knowledge) are disproportionately likely to hold high-risk assets like Futures & Options (F&O) and Cryptocurrency compared to calibrated investors (Fig. 6). This behavioral anomaly underscores why traditional models of rational participation often fail to capture modern retail dynamics.

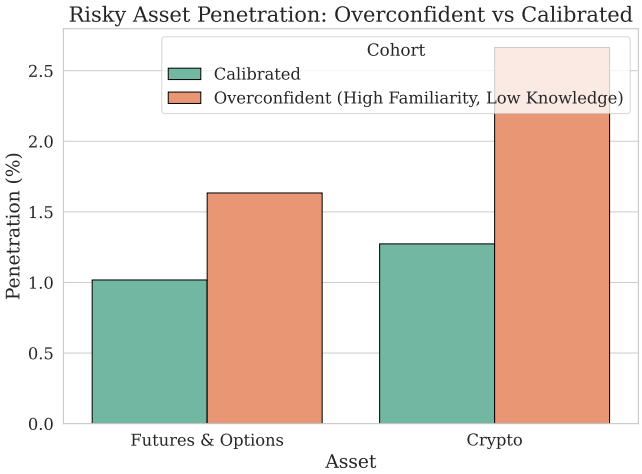

Fig. 6. Risky Asset Penetration. Overconfident investors participate in highly speculative assets at significantly higher rates than calibrated investors.

D. Investor Education Programme (IEP) Exposure

Among the 27,882 active investors, 2,768 (9.9% of all investors) cite SEBI or industry Investor Education Programmes as one of their top information sources.

As shown in Fig. 7, IEP-exposed investors score markedly higher on the nine-item objective financial literacy quiz relative to the broader non-IEP investor population (5.30 vs. 3.75, a 41% gap). Product awareness is near-identical (10.23 vs. 10.23), indicating the knowledge advantage is specific to conceptual depth rather than product breadth. A before-after design would be needed to confirm causality, but the cross-sectional evidence is consistent with IEP deepening financial comprehension among participants.

The divergence in investment philosophy is equally pronounced: IEP-exposed investors are more likely to cite financial goals as a primary motive (30.2% vs. 22.8%) and less likely to seek quick gains (21.7% vs. 23.3%), suggesting that IEP reorients investors toward patient, goal-based capital formation. They also hold fewer Fixed Deposits (49.3% vs. 55.1%), consistent with a modest tilt toward market-linked instruments. Taken together, these findings position IEP as both a knowledge deepening and a philosophical reorientation intervention.

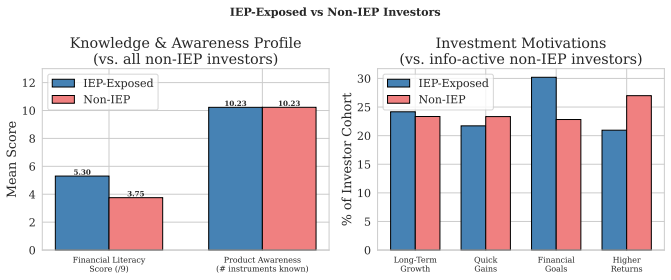

Fig. 7. IEP-exposed vs. non-IEP investors. IEP users score 41% higher on financial literacy; they also show stronger goal orientation and lower quick-gains motivation.

E. Securities Preferences, Demographics, and Influencers

Once households overcome the barrier to entry, their product preferences shift dynamically with income. Fig. 8 illustrates the penetration of various financial instruments across income quartiles. Fixed Deposits (FDs) remain the bedrock of Indian savings, showing uniformly high penetration. However, the adoption of market-linked securities (specifically Mutual Funds/ETFs and Direct Equity) scales dramatically in the higher income brackets, indicating that risk appetite scales non-linearly with wealth.

Crucially, the choice of securities is heavily modulated by where investors source their information. Contrasting investors who rely exclusively on Social Media "Finfluencers" against those using traditional Financial Professionals reveals a sharp behavioral divide. As seen in Fig. 9, the Finfluencer cohort is primarily motivated by "Quick Gains," whereas the professionally-advised cohort seeks "Long Term Growth." (Prevalence here is calculated as the percentage of investors within a specific information-source cohort who selected a given motive). This ties directly to the overconfidence trap discussed earlier, painting a picture of a segmented market where information source dictates risk behavior.

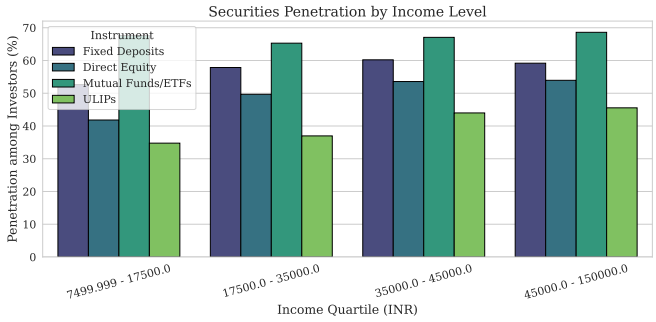

Fig. 8. Securities penetration among investors, segmented by income quartile. Market-linked instruments scale sharply with wealth.

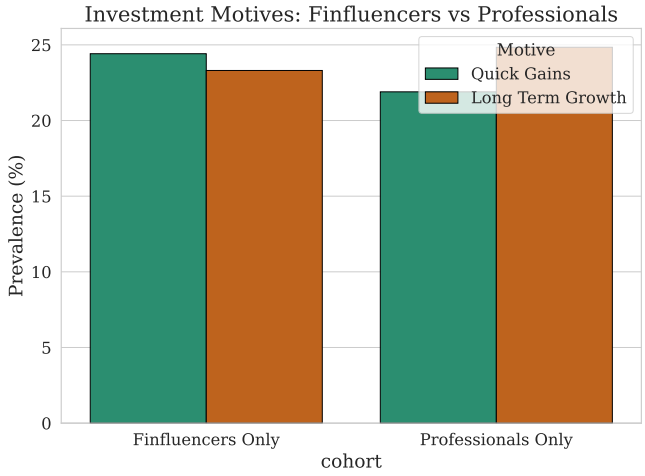

Fig. 9. Investment Motives segmented by Information Source. Prevalence denotes the percentage of investors within each cohort holding a specific motive. Social media heavily biases investors toward short-term speculative gains.

F. Goal–Portfolio Consistency

Beyond demographics and information sources, we test whether household portfolios are internally consistent with their stated primary financial goals. As shown in Fig. 10, households citing long-term objectives such as Retirement (72.4%) and Children's Education (77.2%) exhibit strong alignment, with high penetration in mutual funds and provident fund instruments (PPF/EPF), indicating coherent long-horizon financial planning.

To formalize this, we define a *goal–portfolio consistency* metric as the proportion of households whose primary goal maps to a theoretically aligned instrument set (e.g., Retirement $\rightarrow$ MF/EPF; Daily Trading $\rightarrow$ Equity/F&O) and who hold at least one such instrument.

However, we identify a stark "Say–Do" paradox among speculative entrants. While households citing Daily Trading as their primary goal should, in principle, exhibit high exposure to equities and derivatives, we observe an internal consistency score of just 7.3% (Fig. 11). This is not merely a mismatch but a structural phenomenon: households self-identify with a trading-oriented financial identity without participating in the asset classes that operationalize such behavior.

We therefore classify households into three segments: (i) *aligned investors* (goals and portfolios match), (ii) *constrained investors* (risk-oriented goals but conservative portfolios), and (iii) *aspirational investors* (speculative self-identification without corresponding asset adoption). The "daily trading" cohort overwhelmingly falls into the third category.

This misalignment provides a structural explanation for the overconfidence patterns documented earlier. Aspirational investors exhibit high perceived market familiarity but lack both instrument exposure and technical literacy, creating the precise conditions for Dunning–Kruger effects. In practice, this segment is likely to engage with markets episodically and without adequate risk management, making it particularly vulnerable to adverse financial outcomes.

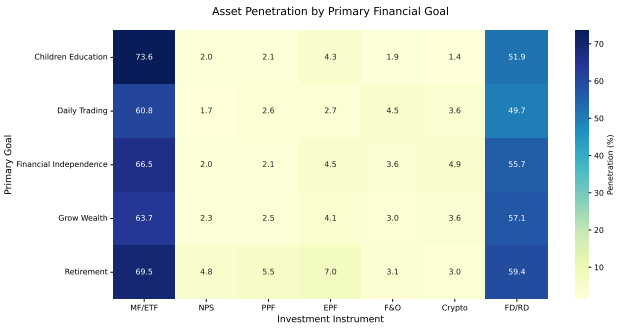

Fig. 10. Asset penetration by primary financial goal. While long-term planners show high adoption of MFs and FDs, the speculative cohort exhibits surprisingly low adoption of their target instruments.

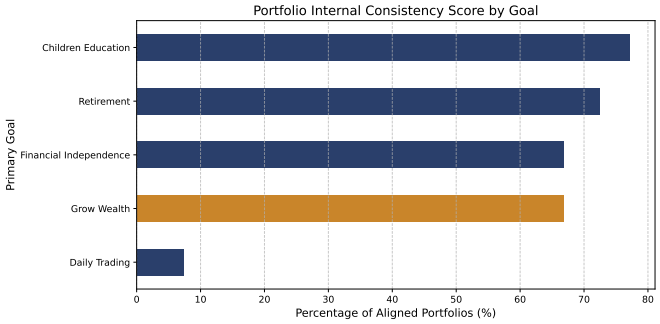

Fig. 11. Internal Consistency Score across primary goals. Consistency is measured as the presence of goal-appropriate instruments in the portfolio. The speculative 'Daily Trading' cohort shows a massive intent–action gap.

G. Investment Horizons

Holding duration preferences (Fig. 12) challenge the notion that all retail investors are short-term speculators. A significant plurality of equity and mutual fund investors define their holding periods as "Long term". In contrast, sophisticated derivative products (F&O) are predominantly utilized for "Short term" horizons, aligning with the motivations of the Finfluencer-driven cohort.

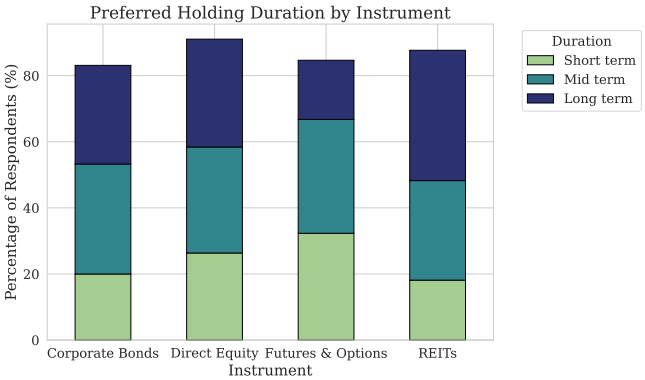

Fig. 12. Preferred holding durations across different asset classes. Direct equities are largely held for the long term, whereas derivatives skew short term.

IV. Predictive Modeling: Forecasting Household Participation

To operationalize our findings, we utilized Histogram-based Gradient Boosting Classifiers to predict household behavior. All three models incorporate product awareness ($n_products_aware$) alongside standard demographic and socioeconomic features. For investor-specific models (Models 2 and 3), the feature space is further expanded to include the behavioral traits uncovered in our earlier analysis: perceived familiarity, objective financial literacy, and information source reliance.

As shown in Fig. 13, the feature importance profiles differ sharply across models: awareness dominates participation prediction, while familiarity and objective knowledge take over once inside the market.

A. Model 1: Predicting Participation

Objective: Predict the binary outcome of whether a household is an active investor (1) or a non-investor (0).
Data Fields Used: Years of Education, Log-transformed Monthly Income, Urban Residency (Binary), Gender (Binary), Risk Tolerance Preference Score, and Product Awareness ($n_products_aware$).
Algorithm & Split: HistGradientBoostingClassifier (200 max iterations, balanced class weights) trained on an 80/20 train-test split.
Performance: The model achieved an AUC of **0.814**.
Insight: Product awareness is the single most predictive feature, consistent with its dominant OR of 2.22 in the logistic

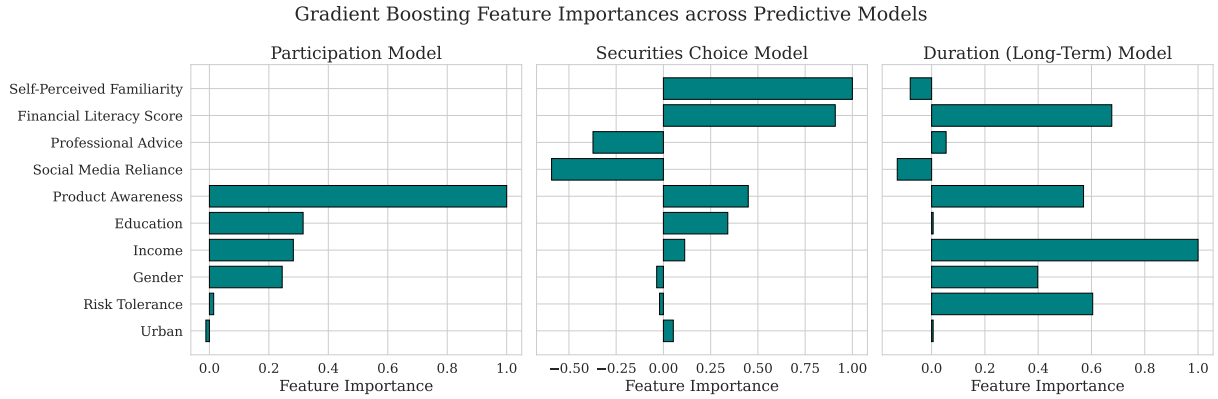

Fig. 13. Normalized feature importances across the three predictive models.

regression. Awareness of more instruments is independently predictive beyond any demographic factor.

B. Model 2: Predicting Securities Choice

Objective: For households that *are* investors, predict if they will hold market-linked securities (Equities or Mutual Funds = 1) versus sticking exclusively to traditional instruments like Fixed Deposits (0).
Data Fields Used: All features from Model 1 (including Product Awareness), **plus** Self-Perceived Stock Market Familiarity, Actual Financial Literacy Score, Social Media Reliance (Binary), and Professional Advice Reliance (Binary).
Algorithm & Split: HistGradientBoostingClassifier trained on an 80/20 train-test split.
Performance: The model achieved an AUC of 0.820.
Insight: Predicting *how* individuals participate requires behavioral context. Once we include perceived familiarity and objective knowledge, the model's discriminatory power surges.

C. Model 3: Predicting Investment Horizon

Objective: For equity investors, predict if they will define their holding period as "Long Term" (1) versus Short/Mid-term (0).
Data Fields Used: Same extended demographic, awareness, and behavioral feature set used in Model 2.
Algorithm & Split: HistGradientBoostingClassifier trained on an 80/20 train-test split.
Performance: The model achieved an AUC of 0.612.
Insight: Forecasting *how long* an investor will hold equity remains the hardest prediction task, suggesting that horizon is shaped by situational factors outside our cross-sectional feature space.

These predictive models validate our core thesis: awareness and behavioral context, not demographics alone, determine participation and post-entry choices. The divergence in feature importance across models underscores that financial outreach is a quantifiably more powerful lever than income redistribution for broadening market access.

V. Conclusion and Policy Implications

The 2025 SEBI Household Survey provides compelling evidence that while the Indian securities market is growing, entry remains fundamentally gated by education and income. Once inside the market, however, traditional demographics fail to predict behavior. Instead, a dangerous cocktail of overconfidence and social media influence is driving a subset of retail investors toward highly speculative, short-term derivative trading.

For policymakers, the mandate is clear: interventions must pivot from merely reducing transaction costs to aggressively policing the quality of financial information. Bridging the knowledge gap via targeted, gamified financial literacy programs is essential. Furthermore, the undeniable Dunning-Kruger effect suggests that access to complex derivatives should perhaps be gated by objective literacy tests, rather than self-reported familiarity, to protect the new wave of retail investors from behavioral pitfalls.

References

[1] SEBI, "Household Survey on Securities Market Participation 2025," Securities and Exchange Board of India, 2025.
[2] Kruger, Justin & Dunning, David. (2000). Unskilled and Unaware of It: How Difficulties in Recognizing One's Own Incompetence Lead to Inflated Self-Assessments. Journal of Personality and Social Psychology. 77. 1121-34. 10.1037//0022-3514.77.6.1121.